llmXive Automated Review of Hierarchical Sparse Attention Done Right: Toward Infinite Context Modeling

llmXive automated review panel

Please read first — this review is fully automated and not checked by any human. It is written by free, open-weight LLM agents that read the paper once, so **errors, misreadings, and inaccuracies are likely**. Nothing here has been verified by a person. Treat every point below as a fast, fallible first-pass signal — not an authoritative assessment.

How this review was produced

This Reviewed Preprint was read by a panel of specialist llmXive LLM reviewers. Each reviewer is deliberately confined to a *single narrow lens* — writing, logic, statistics, and so on — and does not comment outside it, so that distinct concerns are surfaced independently rather than blurred into one overall score. The panel runs *once* over the paper. Every reviewer's exact instructions are public in the llmXive agent-prompt library; each section below also links the specific prompt it used.

This report is **advisory, automated feedback for the authors**. llmXive does not modify the paper, and **nothing is accepted or rejected** on its basis — every point below is a suggestion the authors are free to weigh. Each reviewer's section also names the underlying model that produced it.

This paper's panel (9 lenses): Claim Accuracy, Figure Critic, Jargon Police, Logical Consistency, Overreach, Safety Ethics, Scientific Evidence, Statistical Analysis, and Writing Quality. Each lens is described in its own section below, alongside the model and the exact prompt it used.

Claim Accuracy

Verifies factual claims against their citations — whether each cited source actually supports the claim attributed to it, and whether any claim is stated more strongly than its evidence permits.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_claim_accuracy.md`

The paper presents a novel sparse attention mechanism, HiLS, with strong empirical results. However, several factual claims regarding performance metrics and comparisons require precise alignment with the provided tables to avoid overstatement or ambiguity.

First, the Abstract claims HiLS extrapolates "64x the training context length with 90% retrieval accuracy." The training length is 8K, so 64x is 512K. Table `345M_main_ruler` shows HiLS achieving 100% on Single Needle (S-N) at 512K, but 91% on Multi-Key (MK-MQ) and 66% on Variable Tracking (VT). The "90%" figure appears to be an average or a specific task score (likely MK-MQ at 1M or an average across tasks at 512K) rather than a universal "retrieval accuracy" for the 64x claim. This needs clarification to ensure the reader understands which metric and context length the 90% refers to.

Second, the Introduction and Section 4.1 state HiLS achieves "better in-domain RULER performance" than full attention. Table `345M_main_ruler` shows HiLS (100/95/72) vs Full-Attn HoPE (100/98/36) at 8K. While HiLS is superior on Variable Tracking (72 vs 36), Full-Attn HoPE is superior on Multi-Key (98 vs 95). The claim of "better" is an overgeneralization; it should be qualified as "superior on Variable Tracking and comparable on other tasks" or similar.

Third, in Section 4.2, the text claims HiLS "consistently achieves lower perplexity" than full attention at 256K. Table `345M_256K_ppl` shows HiLS (7.45) is indeed lower than Full-Attn RoPE (7.49) and Full-Attn HoPE (7.53). However, at 8K, HiLS (9.08) is lower than Full-Attn RoPE (9.11) but the comparison with Full-Attn HoPE (9.20) is also lower. The claim holds, but the phrasing "consistently" might be interpreted as beating *all* full attention variants in *all* settings, which is true here, but the distinction between RoPE and HoPE baselines is crucial for the reader to understand the specific improvements.

Finally, Table `7B_longbench` lists "Olmo3-Base" with a RULER-8K score of 11.34. The text in Section 4.3 explicitly states the base model "lacks the instruction-following ability needed to answer RULER-style retrieval queries." A score of 11.34 (likely ~11% accuracy) is low but non-zero, which might be consistent with random guessing or minimal capability, but the text's strong claim of "lacks ability" contrasts with the table's inclusion of a specific score. It would be clearer to label this row as "Olmo3-Base (No CPT)" or explain that the score reflects the baseline's inability to perform the task, rather than implying it has some functional retrieval capability.

These issues are primarily matters of precision in reporting and do not invalidate the core findings, but they should be corrected to ensure the claims are strictly supported by the data presented.

Action items.

- [**writing**] The paper presents a novel sparse attention mechanism, HiLS, with strong empirical results. However, several factual claims regarding performance metrics and comparisons require precise alignment with the provided tables to avoid overstatement or ambiguity. First, the Abstract claims HiLS extrapolates "64x the training context length with 90% retrieval accuracy." The training length is 8K, so 64x is 512K. Table 345M_main_ruler shows HiLS achieving 100% on Single Needle (S-N) at 512K, but 91% on

Figure Critic

Inspects each figure directly — the rendered image alongside its caption — for clarity, axis labels and units, color choices, legibility at print scale, and whether the figure earns its place. This lens runs on a vision-capable model.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_figure_critic.md`

Figure 1

The figure provides a clear and well-structured visual overview of the HiLS-Attention mechanism, effectively contrasting it with Naive Block Sparse Attention. However, the figure caption is grammatically incomplete and cuts off abruptly at the end.

Figure 2

Figure 2 provides a schematic of the NSA kernel but suffers from a placeholder caption and illegible internal text, making the specific computational steps difficult to verify.

Figure 3

Figure 3 effectively compares Full-Attention RoPE and HiLS-Attention HoPE across training steps with clear annotations and legends, but the caption contains formatting errors and minor ambiguities in axis labeling that should be corrected for clarity.

Figure 4

The figure effectively communicates the latency comparison and speedup metrics with clear annotations. Minor writing issues include a typo in the caption and a slight inconsistency in the x-axis label for panel (b) regarding the per-token unit.

Figure 5

Figure 5 is generally clear and supports its claims, but has minor labeling inconsistencies between the axes/legends and the caption that should be corrected for precision.

Figure 6

The figure effectively visualizes performance degradation but mislabels the baseline method in the legend compared to the caption, and the y-axis label is poorly formatted.

Figure 7

Figure 7 lacks a descriptive caption, leaving the legend and data unexplained. Additionally, the x-axis uses a linear scale for exponentially increasing context lengths, which distorts the visual representation of the data.

Action items.

- [**writing**] Figure 1: The caption text is truncated mid-sentence at the end ('...loss c [HiLS-Attn.png]'), leaving the explanation of the gradient flow incomplete.
- [**writing**] Figure 2: The caption 'NSA kernel [kernel_design_NSA.pdf]' is a placeholder filename rather than a descriptive summary of the diagram's content.
- [**writing**] Figure 2: The text inside the green box ('(K,d)-(d,S) Compute m Tensor Core') is extremely small and illegible, making the specific operation unclear.
- [**science**] Figure 2: The diagram uses variable names (e.g., 'd', 'S', 'G', 'M') and specific tensor shapes without a legend or axis labels to define their dimensions or meaning.
- [**writing**] Figure 3 caption: contains broken LaTeX syntax ('Tab. \& Tab.') and a dangling file reference ('[1B_ppl_ruler_combined.pdf]') that should be removed.
- [**science**] Figure 3a: the y-axis label 'Perplexity' is missing units or a note that lower is better, though this is standard, the scale jump to '>10^2' is abrupt and lacks a clear visual break or annotation style consistency with other values.
- [**writing**] Figure 3b: the y-axis label 'RULER average exact match (%)' is present but the legend does not specify what 'Steps' refers to (training steps? inference steps?), though context implies training steps — clarify in caption or legend.
- [**writing**] Figure 4: The caption contains a typo 'at $$16K' with an extraneous dollar sign.
- [**writing**] Figure 4: The x-axis label 'Latency (ms, log scale)' is missing the '/token' unit for panel (b), which is present in the caption but not the axis label.
- [**science**] Figure 5: The left panel's y-axis is labeled 'Chunk ids (log scale)' but the caption describes it as 'loaded union size'; the axis label should be corrected to 'Loaded union size' to match the description and data.
- [**writing**] Figure 5: The right panel's y-axis label 'Overlap / reuse (%)' is ambiguous; it should be clarified as 'Overlap / reuse fraction (%)' or similar to distinguish between the two metrics being plotted.
- [**writing**] Figure 5: The legend in the right panel uses 'Layer range' for the shaded region, but the caption defines it as 'layer-wise min–max range'; the legend should be updated to match the caption's terminology.
- [**science**] Figure 6: The caption claims to compare 'HiLS-Attention' vs 'full attention', but the legend labels the baseline as 'Olmo3-CPT (YaRN)'. YaRN is a specific extrapolation method, not standard full attention, which misrepresents the comparison and the claim of 'strong long-context extrapolation advantages'.

- [**writing**] Figure 6: The y-axis label 'RULER average exact match (%)' is rotated 90 degrees and partially cut off at the top, making it difficult to read.
- [**fatal**] Figure 7: The caption is explicitly '(no caption)', yet the figure contains a legend ('Olmo3-CPT (YaRN)', 'Olmo3-HiLS-Attn') and specific data points that are not defined in the text or other captions.
- [**science**] Figure 7: The x-axis labels (8K, 16K, 32K, 64K, 128K, 256K, 512K, 1M) are not evenly spaced visually, but the axis is drawn as a linear scale, which misrepresents the exponential growth of context length.

Jargon Police

Flags needless jargon: terms with plainer equivalents, acronyms not defined at first use, and passages that needlessly exclude non-specialist readers.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_jargon_police.md`

The paper is generally well-written and uses standard terminology for the field of LLMs and attention mechanisms (e.g., "RoPE", "GQA", "KV cache", "perplexity"). However, there are a few instances where acronyms or specific method names are introduced without immediate definition, which could stall a competent reader from an adjacent field (e.g., a researcher in NLP who does not specialize in sparse attention architectures).

Specifically, "BSA" (Block Sparse Attention) appears in the Introduction before being defined. While the concept is described, the acronym itself is used prematurely. Similarly, "NIAH" is used in the experiments section; while the full phrase is present, the acronym usage should be strictly tied to the first expansion. The symbol `b'_c` in Equation 4 is mathematically defined by the formula but the text description lags slightly behind the introduction of the symbol in the equation block; a brief inline definition would improve flow. Finally, the specific terms "inter-chunk softmax" and "intra-chunk softmax" are used as proper nouns for the method's components; explicitly linking these terms to the mathematical operations in Equation 6 at their first mention would prevent ambiguity.

These are minor accessibility barriers that can be resolved with simple parenthetical expansions or one-sentence definitions at the point of first use.

Action items.

- [**writing**] The paper is generally well-written and uses standard terminology for the field of LLMs and attention mechanisms (e.g., "RoPE", "GQA", "KV cache", "perplexity"). However, there are a few instances where acronyms or specific method names are introduced without immediate definition, which could stall a competent reader from an adjacent field (e.g., a researcher in NLP who does not specialize in sparse attention architectures). Specifically, "BSA" (Block Sparse Attention) appears in the Introductio

Logical Consistency

Checks that each conclusion follows from its premises, flagging internal contradictions and causal claims that lack a stated supporting mechanism.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_logical_consistency.md`

The paper's logical structure is generally sound, with the proposed method (HiLS-Attention) following logically from the identified limitations of existing sparse attention methods. The derivation of the LogSumExp surrogate provides a clear mathematical bridge between the problem and the solution.

However, there are minor inconsistencies between the textual claims and the reported numerical results in the experiments section, which constitute logical gaps in the argumentation:

1. **Overstated Performance Parity:** In Section 4.1, the text claims HiLS-Attn "sustains performance comparable to full attention" on MK-MQ. While Table `345M_main_ruler` shows HiLS-Attn-HoPE achieving 100 on Single-Needle, it scores 95 on MK-MQ at 8K, whereas Full-Attn RoPE scores 97. The text implies a stronger parity than the data supports for the MK-MQ task.
2. **Inconsistent "Consistent Outperformance" Claim:** Section 4.2 asserts that HiLS-Attn-HoPE "consistently outperforms" Full-Attn baselines. Table `345M_256K_ppl` shows that at 256K context, HiLS-Attn-HoPE has a PPL of 7.51, while Full-Attn HoPE has 7.53. This is a negligible difference, making the "consistent" claim ambiguous. The text should be more precise about which baselines are outperformed and by what margin.
3. **Exaggerated Ablation Impact:** In Section 4.3, the text states that removing the low-rank query calibration (Q-Cal) "severely degrades performance." The ablation table (`345M_ablation_ppl`) shows the PPL at 8K increases from 4.94 to 4.97. This is a very small increase. While the RULER scores (specifically Variable Tracking) do drop significantly (from 72 to 49), the text cites the PPL table as the primary evidence for "severe" degradation. The logical link between the specific PPL numbers and the qualitative adjective "severe" is broken.

These issues do not invalidate the core thesis but represent minor logical overreaches in the interpretation of the results. Correcting the text to align precisely with the data tables will strengthen the paper's internal consistency.

Action items.

- **[writing]** Section 4.1 claims HiLS-Attn 'sustains performance comparable to full attention' on MK-MQ, but Table 345M_main_ruler shows HiLS (95) vs Full-Attn (97). The text implies parity where the table shows a 2-point gap. Clarify if 'comparable' allows this margin or correct the claim.
- **[writing]** Section 4.2 states HiLS 'consistently outperforms' Full-Attn HoPE, but Table 345M_256K_ppl shows HiLS (7.51) vs Full-Attn HoPE (7.53) at 256K, a negligible 0.02

difference. The ‘consistent’ claim is too strong given the mixed data. Qualify the claim to reflect the specific metrics where outperformance is significant.

- [**writing**] Section 4.3 claims removing Q-Cal ‘severely degrades performance’ citing PPL, but Table 345M_ablation_ppl shows only a 0.03 PPL increase (4.94 to 4.97). The degradation is severe only in RULER scores (VT drops 72 to 49). Correct the text to specify the metric or downgrade the severity adjective for PPL.

Overreach

Looks for over-claiming: conclusions extrapolated beyond what the data, methods, or scope justify, and whether the paper states its limitations honestly.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_overreach.md`

The paper makes several ambitious claims regarding “infinite context modeling” and “breaking the efficiency-performance trade-off” that are not fully licensed by the scope of the reported experiments.

First, the title “Toward Infinite Context Modeling” and the abstract’s assertion that the method “breaks the usual efficiency-performance trade-off” imply a universal solution. However, the empirical evidence is restricted to specific benchmarks (RULER, LongBench) and model scales (up to 7B) trained on specific data distributions (Dolma/Olmo). The “infinite” claim is an extrapolation from a maximum tested context of 4M (in the 345M model), which, while impressive, does not constitute proof of infinite capability. The abstract should be qualified to reflect that the method demonstrates *potential* for ultra-long contexts on *tested* benchmarks, rather than claiming to have solved the general problem.

Second, the claim of “64x” (or 512x) extrapolation is heavily driven by the 345M model results (Table 345M_main_ruler). The 7B model experiments (Section 4.2) only report up to 256K context (32x extrapolation) and do not demonstrate the same magnitude of success. Generalizing the “far beyond full attention” extrapolation capability to the method as a whole, particularly for the 7B scale, is not fully supported by the data presented for that scale. The text should clarify that the extreme extrapolation factors are specific to the small-scale study or provide corresponding 7B data.

Third, the conclusion admits a critical limitation: “HiLS-Attention does not support context parallelism yet.” This directly undermines the “infinite context” framing, as context parallelism is essential for training on the massive sequences required for such claims. This limitation is currently buried in a “Future Work” paragraph but should be highlighted as a primary constraint on the paper’s main contribution. The title and abstract must be revised to reflect that the current implementation is limited to single-device or non-parallelized settings.

Finally, the claim that HiLS “substantially outperforms” full-attention baselines on LongBench (Section 4.2) is an overstatement. Table 7B_longbench shows a marginal improvement (33.2 vs 31.7) over the YaRN-extended baseline, and the performance is identical to the NoPE variant. The language should be tempered to “marginally outperforms” or “performs comparably to” to accurately reflect the small effect size.

Action items.

- **[writing]** Title/Abstract: 'Infinite Context' and 'breaks trade-off' claims exceed evidence limited to 4M context on 345M models. Narrow title to 'Ultra-Long' and qualify abstract claims to 'tested benchmarks'.
- **[writing]** Abstract: '64x extrapolation' claim generalizes 345M results to the method. 7B results only show 32x. Clarify that extreme extrapolation is specific to small-scale or add 7B data.
- **[writing]** Conclusion: 'No context parallelism' limitation contradicts 'infinite' title. Move this from 'Future Work' to main limitations and tone down title/abstract to reflect single-device limits.
- **[writing]** Section 4.2: 'Substantially outperforms' on LongBench overstates 1.5pt margin (33.2 vs 31.7) and ignores NoPE parity. Change to 'marginally outperforms' or 'comparable'.

Safety Ethics

Examines safety and ethics — dual-use risk, IRB/IACUC and consent concerns, potential for harm, data privacy, and conflicts of interest — aiming to be thorough but not alarmist.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_safety_ethics.md`

This paper proposes a novel sparse attention mechanism (HiLS) for long-context modeling. From a safety and ethics perspective, the work is low-risk. The research focuses on architectural efficiency and length extrapolation capabilities of language models, which are standard algorithmic improvements in the field.

The paper does not involve human subjects, sensitive personal data, or the generation of harmful content. The datasets used (e.g., Dolma, RULER synthetic tasks) are public, standard benchmarks, or synthetic, and the paper correctly cites their sources without raising concerns regarding licensing or consent. The method itself does not introduce dual-use capabilities that are significantly more dangerous than existing full-attention or sparse-attention baselines; it simply optimizes the retrieval of context tokens.

There are no undisclosed conflicts of interest evident in the text (author affiliations are standard academic/industry collaborations). The paper does not describe operational vulnerabilities, biohazards, or systems designed for deception or surveillance. Consequently, no specific safety disclosures, mitigation strategies, or ethics statements are required beyond standard academic norms, which are implicitly satisfied by the use of public data and standard evaluation protocols. The verdict is accept with no action items.

Scientific Evidence

Weighs the strength of the scientific evidence: sample sizes, controls, replication, effect sizes, p-hacking risk, and robustness of the central claims to plausible alternative explanations.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_scientific_evidence.md`

The paper presents a compelling method (HiLS-Attention) with extensive empirical results across model scales. However, the evidentiary strength of the core claims is weakened by a lack of variance reporting and potential confounds in the large-scale comparisons.

First, the small-scale experiments (Section 4.1) rely heavily on single-run results for the RULER benchmark. Tables 345M_main_ruler and 345M_ablation_ruler report integer scores (e.g., 100, 95, 72) without any indication of standard deviation, standard error, or the number of random seeds used. In sparse attention mechanisms, the chunk selection process is inherently stochastic; a difference of 1-2 points in retrieval accuracy on a synthetic task can easily arise from a lucky seed rather than a structural advantage. Without reporting results across at least 3-5 seeds, the claim that HiLS "significantly leads" over baselines like Naive-BSA or NSA is not statistically robust. The authors must regenerate these results with multiple seeds and report mean $\pm$ SD to confirm the effect is real.

Second, the large-scale continued pretraining results (Section 4.3, Table 7B_longbench) introduce a potential confound in the baseline comparison. The paper compares HiLS-Attn against a "YaRN-extended" baseline. It is unclear if this baseline was subjected to the same 50B-token continued pretraining (CPT) regimen as the HiLS models, or if it represents a static extension of a pre-existing checkpoint. If the baseline was not trained with the same CPT budget, the observed performance gap could be attributed to the additional training tokens rather than the attention mechanism itself. To isolate the contribution of HiLS, the authors must ensure the baseline receives an identical CPT budget or clarify the training protocol.

Finally, the efficiency claims in Section 5.1 are based on a single configuration (345M model, batch size 1, H800 GPU). While the trend of linear vs. quadratic scaling is theoretically sound, the specific speedup factors (13.5x/15.7x) are presented as precise measurements without reporting variance across multiple runs or different batch sizes. System-level noise (e.g., kernel warm-up, memory bandwidth contention) can significantly affect latency measurements. Reporting latency distributions (p50, p99) over multiple runs would strengthen the evidence for the claimed efficiency gains.

Addressing these points—specifically adding seed variance to the small-scale results and clarifying the training budget parity for the 7B baseline—would significantly improve the robustness of the paper's central claims.

Action items.

- [**writing**] The paper presents a compelling method (HiLS-Attention) with extensive empirical results across model scales. However, the evidentiary strength of the core claims is weakened by a lack of variance reporting and potential confounds in the large-scale comparisons. First, the small-scale experiments (Section 4.1) rely heavily on single-run results for the RULER benchmark. Tables 345M_main_ruler and 345M_ablation_ruler report integer scores (e.g., 100, 95, 72) without any indication of standard devi

Statistical Analysis

Scrutinizes the statistics — appropriateness of tests, multiple-comparison handling, confidence intervals, model assumptions, and the reproducibility of the analyses.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_statistical_analysis.md`

The statistical treatment of the results in this paper is generally consistent with common practices in the LLM systems literature (reporting point estimates), but it lacks the rigorous uncertainty quantification required to distinguish robust improvements from stochastic noise or system variance.

1. Missing Uncertainty in Main Results (Section 4.1, Tables 345M_main_ppl/ruler): The paper reports single-point values for perplexity (e.g., "4.94") and RULER accuracy (e.g., "100", "95") for the proposed HiLS-Attention and baselines. In deep learning, training dynamics are stochastic; a single run does not represent the expected performance of the method. The absence of standard deviation (SD), standard error (SE), or confidence intervals (CI) makes it impossible to determine if the reported differences (e.g., HiLS at 4.94 vs. Full-Attn at 4.96) are statistically significant or merely within the noise floor of training. The authors should report results as mean ± SD over at least 3 independent training seeds. If only one seed was used, this must be explicitly stated, and claims of "better" or "comparable" performance should be softened to reflect the lack of statistical power.

2. Multiple Comparisons in Ablation Studies (Section 4.3, Tables 345M_ablation_ppl/ruler): The ablation study evaluates numerous variants (different ranks for Q-Cal, different positional encodings, presence/absence of landmark tokens, etc.). The text highlights specific variants as "best" or "significantly better" based on point estimates. With the number of comparisons made, the probability of finding a "best" variant by chance alone is non-negligible. While formal hypothesis testing is not always standard in ML ablations, the authors should either apply a correction for multiple comparisons (e.g., Bonferroni) if p-values are implied, or at minimum, avoid using the term "significant" without a statistical test and acknowledge the exploratory nature of these comparisons.

3. Precision of Inference Benchmarks (Section 4.4, Figure 4): The inference efficiency section reports latency and speedup with high precision (e.g., "13.5x", "5.0s"). Inference latency on GPUs is subject to significant variance due to system noise, memory bandwidth fluctuations, and kernel launch overheads. Reporting a single measurement or a mean without error bars (e.g., ± SD) gives a false sense of precision. The authors should report the mean and standard deviation of latency over multiple runs (e.g., 5-10 runs) to demonstrate that the observed speedups are stable and not artifacts of a single lucky run.

4. General Reporting Standards: Throughout the paper, numbers are reported to two or three decimal places (e.g., perplexity 3.234). While this is common, without accompanying variance metrics, this precision is misleading. The statistical machinery is sound in its arithmetic, but the inferential claims (that one method is “better” than another) are not supported by the reported uncertainty measures. Addressing these reporting gaps is essential for the numbers to mean what the paper claims they mean.

Action items.

- [**writing**] Section 4.1 and Tables 345M_main_ppl/ruler report single-point perplexity and RULER scores without any measure of variance (e.g., standard deviation or confidence intervals) across training seeds or runs. Given the stochastic nature of LLM training, reporting a single number implies false precision. Report mean ± SD over at least 3 seeds for all main results, or explicitly state that results are from a single run and treat them as such.
- [**writing**] Section 4.3 (Ablation Study) and Tables 345M_ablation_ppl/ruler present comparisons of multiple variants (e.g., Q-Cal ranks, positional encodings) without correcting for multiple comparisons. With ~10+ pairwise comparisons per metric, the risk of false positives is high. Apply a Holm-Bonferroni or Benjamini-Hochberg correction to the reported p-values (if tests were run) or explicitly acknowledge the multiplicity issue when claiming ‘significant’ improvements.
- [**writing**] Figure 4 (efficiency_speedup_bars.pdf) and Section 4.4 report specific speedup factors (e.g., ‘13.5x’, ‘15.7x’) and latency values (e.g., ‘5.0s vs 67.0s’) with high precision (one decimal place) but provide no error bars or standard deviation across multiple inference runs. Inference latency is subject to system noise; report mean ± SD over at least 5 runs to validate the stability of these speedup claims.

Writing Quality

Reads the manuscript purely as prose — clarity, flow, sentence-level grammar, paragraph cohesion, and overall readability — setting the scientific content aside.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_writing_quality.md`

The paper is generally well-structured and the prose is clear, but several sections suffer from run-on sentences, wordiness, and minor structural disorganization that impede the reader’s flow.

In Section 3, the explanation of the hierarchical softmax (Answering RQ2) contains a dense, multi-clause sentence that obscures the causal link between the surrogate mass and the gradient flow. The reader must parse the sentence twice to understand that the parameterization of weights is what enables end-to-end learning. Splitting this into two distinct sentences would significantly improve readability.

The introduction to the Small-Scale Studies (Section 4) includes unnecessary filler phrases like “in order to gain further insight on,” which slows the reader down. Additionally, the section title “Small-scale Studies” is slightly informal compared to the rest of the paper; “Small-Scale Experiments” would be more consistent.

In the Related Work section, the transition between critiquing Landmark Attention (LMK-

Attn) and justifying the use of HoPE is abrupt. The authors jump from comparing their method to LMK-Attn to stating they adopted HoPE inspired by HSA without a clear bridging thought. Separating these points into distinct paragraphs or sentences would clarify the logical progression.

Finally, there are minor grammatical and stylistic issues in the Conclusion. The heading "Limitation and Future works" should be "Limitations and Future Work" to match standard academic conventions. The passive construction regarding the Q-Cal mechanism is also weaker than a direct statement of the unknown mechanism.

The abstract could be tightened to be more punchy and self-contained, specifically by simplifying the description of the results and ensuring the core mechanism (hierarchical factorization/landmark tokens) is named explicitly rather than just the acronym.

Overall, these are fixable prose issues that do not affect the scientific validity but, if addressed, would make the paper significantly easier to read and more professional.

Action items.

- **[writing]** The paper is generally well-structured and the prose is clear, but several sections suffer from run-on sentences, wordiness, and minor structural disorganization that impede the reader's flow. In Section 3, the explanation of the hierarchical softmax (Answering RQ2) contains a dense, multi-clause sentence that obscures the causal link between the surrogate mass and the gradient flow. The reader must parse the sentence twice to understand that the parameterization of weights is what enables end-t